Deep Learning Project : Review of Learning Transferable Visual Models From Natural Language Supervision

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Abstract

Contrastive Language–Image Pretraining (CLIP) demonstrates that large-scale alignment between images and natural language yields highly transferable visual representations. In this work, we provide a structured review of the CLIP framework, analyze its architectural and objective design, and empirically examine its claims through controlled reproduction experiments. We first evaluate zero-shot transfer and small-scale contrastive training to assess the role of scale. We then conduct linear probing experiments across multiple benchmarks to isolate representation quality under frozen encoders, confirming the strong linear separability of CLIP features. Finally, we extend the contrastive alignment principle to a molecular graph–text setting, demonstrating that the objective generalizes beyond natural images to structured scientific domains. Together, these analyses support the view that multimodal contrastive learning constitutes a general representation paradigm rather than a vision-specific technique. Code is available : https://github.com/fotisk07/Deep-Learning/tree/master/clip-reproduction

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1 Introduction

Modern computer vision systems traditionally rely on supervised learning with manually annotated datasets such as ImageNet. While this paradigm has led to significant progress, it suffers from two structural limitations: (i) fixed label spaces restrict transfer to new tasks, and (ii) dataset annotation is costly and inherently limited in semantic diversity.

Contrastive Language–Image Pretraining (CLIP) [4] proposes a different paradigm: instead of learning from curated class labels, models are trained on large-scale image–text pairs collected from the web. Natural language serves as a flexible and expressive supervision signal, enabling the model to align visual and textual representations in a shared embedding space.

The key idea is simple but powerful. Given a dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N},$$

where x_i is an image and y_i its associated text, CLIP learns two encoders that map images and text into a common d-dimensional space. Training is performed using a contrastive objective that encourages matched image–text pairs to have high similarity while pushing apart mismatched pairs within a minibatch.

Unlike traditional supervised classifiers, CLIP does not learn a fixed output layer tied to a predefined taxonomy. Instead, downstream classification can be performed in a *zero-shot* manner by embedding candidate textual descriptions and selecting the one most similar to the image representation. This shifts the paradigm from task-specific training to general multimodal representation learning.

A central contribution of CLIP lies in scale. The model is trained on approximately 400 million image–text pairs, demonstrating that contrastive multimodal learning benefits significantly from large-scale noisy web data. This scaling strategy, combined with a simple symmetric contrastive loss, yields representations that transfer competitively across a wide range of vision benchmarks.

In this report, we review the methodological foundations of CLIP, analyze its architectural and training design choices, and reproduce a subset of its evaluation protocols. We further investigate representation quality through linear probing and discuss extensions to molecule–text retrieval.

2 Background: Contrastive Representation Learning

Contrastive learning aims to learn representations by distinguishing between similar (positive) and dissimilar (negative) pairs. Instead of predicting explicit labels, the objective enforces invariances and relational structure in the embedding space.

Let $z_i \in \mathbb{R}^d$ denote the representation of sample i. In its general form, contrastive learning optimizes a similarity function $s(z_i, z_j)$ such that positive pairs have high similarity and negatives have low similarity. A widely used formulation is the InfoNCE objective:

$$\mathcal{L}_i = -\log \frac{\exp(s(z_i, z_i^+)/\tau)}{\sum_{j=1}^{B} \exp(s(z_i, z_j)/\tau)},$$

where:

- z_i^+ is a positive sample associated with z_i,
- $\tau > 0$ is a temperature parameter,

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- the denominator runs over samples in the minibatch.

This loss can be interpreted as a softmax classification problem in which each example must identify its correct positive among B candidates. The temperature parameter controls concentration of the distribution and influences gradient sharpness.

In computer vision, contrastive learning gained prominence through self-supervised methods such as SimCLR and MoCo, where positive pairs are generated via data augmentations of the same image. These methods demonstrated that large-scale instance discrimination can produce semantically meaningful features without labels.

CLIP extends this framework to the multimodal setting. Instead of augmentations of a single image, positives consist of aligned image–text pairs. The supervision signal is therefore semantic rather than purely visual. The task becomes cross-modal matching: given an image embedding, identify its corresponding text embedding among a set of candidates, and vice versa.

This shift has important consequences. First, language provides a far richer supervisory signal than fixed class labels. Second, the embedding space acquires semantic structure that supports zero-shot transfer. Finally, contrastive learning becomes a bridge between modalities, enabling unified multimodal representations.

3 The CLIP Method

CLIP learns a joint embedding space for images and text through large-scale contrastive training. The method combines dual encoders, a symmetric contrastive objective, and web-scale supervision. We detail its components below.

3.1 Dual-Encoder Architecture

CLIP consists of two separate neural networks:

- An image encoder $f_\theta : \mathcal{X} \to \mathbb{R}^d$
- A text encoder $g_\phi : \mathcal{T} \to \mathbb{R}^d$

Each encoder maps its modality into a shared d-dimensional embedding space. The outputs are ℓ_2-normalized:

$$v = \frac{f_\theta(x)}{\|f_\theta(x)\|_2}, \qquad t = \frac{g_\phi(y)}{\|g_\phi(y)\|_2}.$$

Image Encoder. Two backbone families are explored in the original paper:

- Modified ResNet [2] architectures with attention pooling replacing global average pooling.
- Vision Transformers (ViT) [1], where images are split into patches and processed via self-attention.

In the ViT configuration, an image is decomposed into fixed-size patches, linearly projected, and processed by a standard Transformer encoder. The representation associated with the special classification token is used as the image embedding.

Text Encoder. The text encoder is a Transformer model [6] operating on byte-pair encoded tokens [5]. It uses masked self-attention, similar to GPT-style architectures. The embedding corresponding to the end-of-text token is used as the sentence representation.

Both encoders are followed by learned linear projection layers mapping into the shared embedding space.

3.2 Symmetric Contrastive Objective

Given a minibatch of B image–text pairs $\{(x_i, y_i)\}_{i=1}^B$, embeddings are computed as:

$$v_i = f_\theta(x_i), \qquad t_i = g_\phi(y_i).$$

The similarity matrix is defined via scaled cosine similarity:

$$s_{ij} = \frac{v_i^\top t_j}{\tau},$$

where $\tau > 0$ is a learnable temperature parameter.

Training minimizes a symmetric cross-entropy loss:

$$\mathcal{L} = \frac{1}{2}\left(\frac{1}{B}\sum_{i=1}^{B} -\log \frac{\exp(s_{ii})}{\sum_{j=1}^{B}\exp(s_{ij})} + \frac{1}{B}\sum_{i=1}^{B} -\log \frac{\exp(s_{ii})}{\sum_{j=1}^{B}\exp(s_{ji})}\right).$$

The first term corresponds to image-to-text matching, while the second corresponds to text-to-image matching. This formulation treats every other element in the minibatch as a negative sample.

The objective can be interpreted as learning a shared metric space where aligned pairs maximize mutual similarity and mismatched pairs are uniformly repelled.

3.3 Zero-Shot Classification

A defining feature of CLIP is its zero-shot evaluation protocol.

Given a downstream classification task with classes $\{c_1, \ldots, c_K\}$, each class name is embedded via a textual prompt, e.g.,

$$t_k = g_\phi(\text{"a photo of a } c_k\text{"}).$$

For a test image x, prediction is performed by computing cosine similarities:

$$\hat{y} = \arg\max_k \frac{f_\theta(x)^\top t_k}{\|f_\theta(x)\|_2 \|t_k\|_2}.$$

No gradient updates are performed. The classifier is implicitly defined by language.

This procedure removes the need for task-specific training and allows direct transfer across diverse benchmarks. Performance depends on prompt design, which introduces a mild form of task adaptation without parameter updates.

3.4 Scaling Strategy

A central contribution of CLIP lies in scale rather than architectural novelty. The model is trained on approximately 400 million image–text pairs collected from the web. The dataset is weakly filtered but remains noisy and diverse.

Scaling both model capacity and dataset size yields consistent improvements in zero-shot performance across benchmarks. Unlike supervised ImageNet training, where label space is fixed, CLIP benefits from the open-ended nature of language supervision.

Scaling model size and dataset size yields consistent improvements in zero-shot transfer. This empirical behavior aligns with broader observations on scaling laws in deep learning [3].

4 Reproduction: A Small-Scale Validation

The objective of this reproduction is not to match the large-scale performance reported in [4], which relies on 400M image–text pairs and substantial computational resources. Instead, our goal is twofold:

(1) To verify the zero-shot behavior of the pretrained CLIP model under distribution shift.
(2) To validate that contrastive language–image training can learn meaningful representations in simplified, small-scale settings.

We emphasize that this section should be interpreted as a controlled toy validation of the framework rather than a competitive benchmark study.

4.1 Evaluation Protocol

We evaluate three different settings:

- **Pretrained CLIP (zero-shot).** We use the publicly available OpenAI CLIP model [4] without any fine-tuning. Classification is performed via cosine similarity between image embeddings and prompt-based class text embeddings. No gradient updates are applied.
- **Supervised ResNet-50 baseline.** A ResNet-50 model [2] is fine-tuned end-to-end on each dataset using labeled supervision. This provides an upper bound for task-specific supervised training.
- **Small-scale CLIP training.** We implement and train our own CLIP variants using significantly smaller datasets. Two configurations are explored: (i) a Vision Transformer (ViT) image encoder [1] paired with a Transformer text encoder [6], closely following the original design; (ii) a ResNet-50 image encoder variant.

4.2 Datasets

We consider MNIST, CIFAR-100, and FashionMNIST. These datasets are intentionally simple and differ substantially from CLIP's large-scale web pretraining distribution. In particular:

- MNIST consists of grayscale handwritten digits.
- FashionMNIST contains grayscale clothing images.
- CIFAR-100 contains natural images with fine-grained object categories.

These choices allow us to probe generalization under varying levels of distribution shift.

4.3 Results

Table 1 reports top-1 classification accuracy.

Table 1: Accuracy (%) comparison.

Model	MNIST	CIFAR100	FMNIST
CLIP (Zero Shot)	29	65	63
ResNet-50 (Fine Tuned)	95	61	79
CLIP (ViT)	65	42	61
CLIP (RN50)	55	49	54

4.4 Interpretation

Several observations can be made.

First, zero-shot CLIP performs competitively on CIFAR-100 despite no task-specific training, confirming the transfer capability emphasized in [4]. However, performance drops significantly on MNIST. This is expected: handwritten digits are far from the natural image distribution of web data, and the semantic alignment between digits and descriptive language is limited.

Second, supervised fine-tuning of ResNet-50 outperforms zero-shot CLIP on MNIST and FashionMNIST. This highlights a key distinction: when abundant labeled data is available for a specific task, end-to-end supervised training remains highly effective.

Third, small-scale CLIP training yields moderate performance but does not approach pretrained CLIP results. This gap illustrates the importance of scale in multimodal contrastive learning. Without massive image–text corpora, the learned embedding space lacks the semantic richness observed in the original work.

Overall, this reproduction confirms two central aspects of CLIP: (i) zero-shot transfer is viable across natural image datasets, (ii) performance strongly depends on large-scale language supervision.

5 Linear Probing Evaluation

5.1 Motivation

Zero-shot evaluation demonstrates that CLIP can perform classification without task-specific training by leveraging natural language prompts. However, zero-shot performance depends on prompt design and text–image alignment quality, which may obscure the intrinsic structure of the learned visual representation.

To isolate the quality of the visual embedding space itself, we follow the linear evaluation protocol adopted in [4]. Linear probing evaluates whether downstream tasks become linearly separable in the frozen feature space induced by the pretrained encoder.

Formally, given a pretrained visual encoder

$$f_\theta : \mathcal{X} \to \mathbb{R}^d,$$

we extract fixed image representations

$$z_i = f_\theta(x_i),$$

and train a linear classifier on $\{z_i\}$ while keeping θ frozen.

This protocol removes the confounding effect of end-to-end fine-tuning and directly probes the geometric organization of the embedding space. If a simple linear decision boundary achieves high accuracy, this indicates that semantic information has been encoded in a linearly accessible form.

Linear probing therefore complements zero-shot evaluation: while zero-shot measures cross-modal alignment, linear probing measures unimodal representation quality.

5.2 Experimental Setup

Datasets. Folliwng the CLIP paper, we evaluate on multiple benchmarks of varying granularity and domain shift:

- **MNIST**: handwritten digits (10 classes).
- **CIFAR-10**: natural object classification (10 classes).
- **CIFAR-100**: fine-grained natural images (100 classes).
- **EuroSAT**: satellite land-use classification.
- **Food-101**: fine-grained food recognition.

These datasets range from simple digit recognition to fine-grained semantic categorization, enabling a diverse evaluation of transfer behavior.

Encoders. We compare:

- The pretrained OpenAI CLIP visual encoder [4].
- A supervised ResNet-50 pretrained on ImageNet [2].

Both encoders are kept frozen during evaluation.

Feature Extraction. For each dataset, features are extracted in evaluation mode with gradients disabled:

$$z = f_\theta(x).$$

Following common practice in contrastive learning, features are ℓ_2-normalized:

$$\tilde{z} = \frac{z}{\|z\|_2}.$$

This ensures that classification depends primarily on angular similarity, consistent with CLIP's cosine-based training objective.

Linear Classifier. On top of frozen features, we train a multinomial logistic regression classifier:

$$p(y = k \mid \tilde{z}) = \frac{\exp(w_k^\top \tilde{z})}{\sum_{j=1}^{K} \exp(w_j^\top \tilde{z})}.$$

The optimization problem is:

$$\min_W -\sum_{i=1}^{N} \log p(y_i \mid \tilde{z}_i) + \lambda \|W\|_2^2.$$

We use L-BFGS with a fixed regularization strength and no hyperparameter search, in order to maintain a controlled and reproducible protocol.

Evaluation metric is top-1 classification accuracy on the test split.

5.3 Results

Table 2 reports linear probing accuracy for our reproduced CLIP encoder, the corresponding results reported in [4] (ViT-B/32), and a supervised ImageNet-pretrained ResNet-50 baseline.

Table 2: Linear probing accuracy (%) comparison. CLIP (paper) corresponds to ViT-B/32 results reported in [4].

Dataset	CLIP (ours)	CLIP (paper)	ResNet-50
MNIST	96.77	99.0	95.76
CIFAR-10	95.09	95.1	91.53
CIFAR-100	78.38	80.5	73.27
EuroSAT	92.69	97.0	94.06
Food-101	91.93	88.8	59.14

Our reproduced results closely follow the qualitative trends reported in [4]. While minor numerical discrepancies are expected due to differences in preprocessing, feature normalization, and the absence of hyperparameter tuning, the relative ranking across datasets is consistent with the original findings.

In particular, CLIP representations remain competitive with or superior to supervised ImageNet features under linear evaluation.

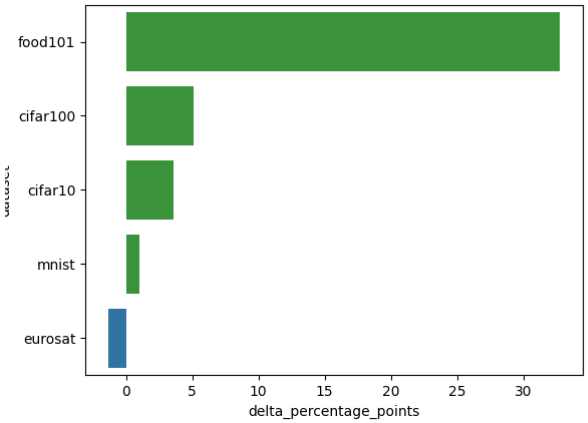

Figure 1: Our reproduced linear probing comparison (CLIP minus ResNet-50 accuracy in percentage points). Positive values indicate stronger linear separability of CLIP features.

The strong performance on CIFAR-100 and Food-101 highlights the benefit of language supervision for semantically rich and fine-grained recognition tasks.

Overall, we recover the same structural conclusion as the original paper: contrastive language–image pretraining produces visual embeddings that are both transferable and linearly separable across diverse downstream benchmarks.

5.4 Discussion

Linear probing provides a direct measure of representation geometry. Since the encoder remains frozen, improvements in accuracy reflect the intrinsic organization of the embedding space rather than downstream adaptation.

The results highlight three key observations:

(1) Contrastive language–image supervision produces highly transferable visual features.
(2) CLIP embeddings often surpass supervised ImageNet representations under linear evaluation.
(3) Large-scale multimodal training induces semantic structure that generalizes across domains.

Taken together with zero-shot evaluation, these findings reinforce the conclusion that CLIP's training paradigm yields representations that are both multimodally aligned and unimodally discriminative.

6 Extensions of the Contrastive Learning Framework

A central claim of CLIP [4] is that contrastive alignment between structured inputs and natural language yields transferable representations. Since the objective is modality-agnostic, it can in principle be applied beyond images to other structured scientific domains.

Molecular graphs provide a particularly relevant test case. Chemical datasets often contain structured graph representations alongside textual annotations describing functional groups, atom counts,

or structural properties. Unlike web-scale vision–language corpora, such datasets are limited in size and exhibit strong semantic redundancy. Extending CLIP to this domain therefore tests whether multimodal contrastive learning remains effective under low-data and high-similarity conditions.

We consider a dataset

$$\mathcal{D} = \{(G_i, y_i)\}_{i=1}^{N},$$

where G_i denotes a molecular graph and y_i a corresponding textual description.

Each graph $G_i = (V_i, E_i)$ encodes atomic features at the node level (atom type, aromaticity, ring membership, etc.) and bond features at the edge level (bond type, conjugation, etc.). The associated text descriptions summarize structural properties, functional groups, and atom counts. The dataset is moderately sized compared to web-scale vision–language corpora and exhibits strong semantic redundancy across captions.

6.1 Architectural Adaptation

Adapting CLIP to this setting requires replacing the image encoder with a graph encoder while preserving the contrastive alignment principle.

Graph encoder. The encoder $h_\theta(\cdot)$ is implemented as a message-passing neural network. Node and edge categorical features are first embedded into dense vectors. A stack of K GNN layers (e.g., GINE convolutions) propagates information along bonds:

$$\mathbf{h}_i^{(k+1)} = \mathrm{MLP}^{(k)}\left(\mathbf{h}_i^{(k)} + \sum_{j \in \mathcal{N}(i)} \left(\mathbf{h}_j^{(k)} + \mathbf{e}_{ij}\right)\right).$$

After message passing, attention pooling produces a fixed-dimensional graph representation. Global handcrafted features (e.g., atom counts or aromatic ratios) are concatenated before a final projection layer maps the representation to the shared embedding space.

Text encoder. The text encoder $g_\phi(\cdot)$ is initialized from a pre-trained language model. Given the relatively small size of the dataset, freezing the backbone and learning only a lightweight projection head often stabilizes training and reduces overfitting. The projection head maps text embeddings into the same d-dimensional space as graph embeddings.

Contrastive Training. Training uses the same symmetric contrastive objective as CLIP, aligning matched graph–text pairs while treating other batch elements as negatives.

6.2 Training Dynamics and Design Choices

Several aspects differ from the original CLIP regime:

Limited scale. Unlike CLIP's large-scale web data, molecular datasets are relatively small. This increases the risk of overfitting and reduces negative diversity. Larger effective batch sizes (via gradient accumulation) partially mitigate this limitation.

Semantic proximity of negatives. Many molecules share similar structural motifs and textual templates. As a result, negatives are often chemically related and semantically close. The contrastive objective must therefore learn fine-grained distinctions rather than coarse modality alignment.

Temperature sensitivity. The temperature parameter τ plays a critical role in preventing representation collapse. Too small a temperature over-concentrates similarity distributions and can amplify noise; too large a temperature weakens discrimination. Empirically, careful tuning is required in smaller scientific datasets.

Projection head capacity. Increasing projection head depth may improve optimization flexibility, but excessive capacity can destabilize alignment in low-data regimes. A two-layer MLP often provides a good trade-off between expressivity and stability.

6.3 Empirical Results of Graph–Text Contrastive Alignment

We evaluate the effectiveness of the adapted CLIP framework on a molecular graph–text dataset included with the code through Recall@K (R@1, R@5, R@10) on a held-out validation split.

Retrieval Performance. The final model achieves strong alignment between molecular graphs and textual descriptions. In particular, Recall@10 approaches saturation (approximately 0.97), indicating that the correct description is almost always contained within the top-10 retrieved candidates. Recall@1 reaches approximately 0.66, reflecting the difficulty of fine-grained discrimination among highly similar captions.

The gap between Recall@1 and Recall@10 is consistent with the structural properties of the dataset: many captions share nearly identical templates and differ only in small numerical details or localized structural features. Consequently, the contrastive objective successfully clusters semantically related graph–text pairs, while exact top-1 separation remains more challenging.

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